

Problem Statement

1. Background and Context

Agriculture is an important source of livelihood for millions of smallholder farmers, particularly in developing countries. Farmers regularly seek advice on crop production, pest and disease management, soil fertility, fertilizer application, irrigation, and climate adaptation to improve their productivity. Agricultural extension workers also depend on reliable and up-to-date information to provide farmers with practical recommendations that support informed decision-making.

Much of this knowledge already exists in agricultural extension manuals, research publications, government guidelines, and other advisory documents. However, these resources are often stored in large document collections, making it difficult to quickly identify the documents that best answer a farmer's question. Since Retrieval-Augmented Generation (RAG) systems rely on retrieving relevant documents before generating responses, the quality of document retrieval directly affects the accuracy and reliability of the advice provided.

2. Core Problem

Farmers and agricultural experts often describe the same problem using different language. Farmers usually ask questions in simple, everyday language, while agricultural extension documents commonly use technical terms and scientific expressions. Although they may describe the same issue, the difference in wording can make it difficult for retrieval systems to recognise the relationship between a farmer's question and the most relevant documents.

For example, a farmer may ask, "Why are my crop leaves turning yellow?" while an extension document discusses "nitrogen deficiency causing chlorosis in maize." Although both describe the same condition, a retrieval system that relies mainly on keyword matching may fail to identify the connection.

This challenge becomes more significant when searching through large collections of agricultural documents. Instead of retrieving the most relevant documents, the system may return documents that contain similar keywords but provide little useful information. Poor document retrieval can therefore reduce the quality of the evidence supplied to a RAG system, leading to responses that are incomplete, inaccurate, or weakly supported.

3. Current Approach and Limitations

The baseline retrieval method considered in this study is Term Frequency–Inverse Document Frequency (TF-IDF), a widely used lexical retrieval technique that ranks documents based on the importance of words shared between a user's query and the documents.

Although TF-IDF is simple and efficient, it relies heavily on exact keyword matching. As a result, it may not recognise synonyms, related agricultural concepts, or different expressions that describe the same problem. It may also perform poorly when farmer questions are short, conversational, or use terminology that differs from the wording found in extension documents. Consequently, the most relevant documents may not always appear among the top-ranked search results, reducing the effectiveness of the RAG system.

4. Research Aim

The aim of this study is to develop and evaluate a document retrieval approach that improves the ranking of relevant agricultural extension documents for smallholder farmers' questions.

The study will compare the TF-IDF baseline with BM25, Sentence Transformer-based retrieval, and a hybrid retrieval approach that combines lexical and semantic retrieval techniques. The study seeks to determine whether these approaches can improve the ranking of relevant documents within the top five retrieval results, thereby providing stronger supporting evidence for Retrieval-Augmented Generation systems.

5. Proposed Technical Approach

This study will compare different document retrieval methods to determine which approach retrieves the most relevant agricultural extension documents for farmers' questions.

The study will begin with TF-IDF as the baseline retrieval method. BM25 will then be evaluated as an improved sparse retrieval model because it generally provides more effective document ranking than TF-IDF.

To capture the meaning of both farmer questions and extension documents, Sentence Transformer embeddings will be used for dense retrieval. Unlike keyword-based methods, dense retrieval can identify documents that are semantically related even when different words or expressions are used.

Finally, a hybrid retrieval approach that combines BM25 and Sentence Transformer embeddings will be implemented to take advantage of both lexical matching and semantic similarity. The performance of these retrieval methods will then be compared to determine which approach provides the most relevant document rankings.

6. Scope and Boundaries

This study focuses on the document retrieval stage of a Retrieval-Augmented Generation (RAG) system. It uses the agricultural extension document collection and farmer questions provided in the Kaggle competition dataset.

The study compares the performance of TF-IDF, BM25, Sentence Transformer retrieval, and a hybrid retrieval approach. The evaluation is limited to document retrieval and ranking. It does not assess or compare the quality of the final responses generated by a language model.

7. Evaluation Method

The retrieval methods will be evaluated using Normalized Discounted Cumulative Gain at rank 5 ($nDCG@5$), the evaluation metric specified for the Kaggle competition. This metric measures how effectively relevant documents are ranked within the top five retrieval results.

The performance of BM25, Sentence Transformer retrieval, and the hybrid retrieval approach will be compared with the TF-IDF baseline to determine whether they improve the retrieval of relevant agricultural extension documents.

8. Significance of the Study

Accurate document retrieval is an important part of any Retrieval-Augmented Generation system because the quality of the retrieved documents influences the quality of the final response. Improving document retrieval can help ensure that farmers and agricultural extension workers have access to more relevant and reliable information when seeking agricultural advice.

The findings of this study may contribute to the development of more effective agricultural advisory systems by improving access to information on crop management, pest and disease control, soil fertility, fertilizer application, and other farming practices. Better retrieval performance can also improve the quality of evidence available to downstream RAG systems, leading to more reliable and well-supported recommendations.